F Training only

Stage 1 · Cross-entropy training

TRAIN reference caption

Shift right

Current decoder

Predicted token distribution

Target tokens

LCE

Stage 2 · SCST + cross-entropy

Current model · same image

Sampled caption y_s

Greedy caption yg

TRAIN references

CIDEr reward r(·)

Current-batch LCE
TRAIN teacher forcing

Advantage: $r(y_s) - r(yg)$

Rollout log $p(y_s)$

LSCST · detached advantage

Total loss = LCE + 0.05 LSCST

References: training and evaluation only.